EXACT PERIODICITY DOES NOT EVADE THE BOLTZMANN BRAIN PROBLEM

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Abstract. Carroll, Diachenko and Dulani proposed that a finite quantum universe with commensurable energy levels evolves periodically, and that its low-entropy bounce then recurs far more often than the law $e^{-\Delta S}$ allows, so that ordinary observers outnumber Boltzmann brains. We prove that this is not so. Born-weighted rates do not depend on whether the spectrum is commensurable. For macrostates in general position, the minimal macrostate carries weight $(d+1)/(D+1)$, and for all but a small fraction of placements the period obeys $T\,\Delta E \gtrsim (D+1)/(d+1)$. The proposed range of periods is therefore empty unless the macrostates are fine-tuned or the dynamics explores no more entropy than a single Boltzmann brain.

Keywords: quantum recurrence; Boltzmann brains; diagonal ensemble.

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1. Introduction

If the universe is a pure state $\psi(t)$ evolving unitarily in a Hilbert space of finite dimension D, its evolution is recurrent [5]. Recurrence brings the Boltzmann brain problem [8, 1]. A system that spends most of its time near equilibrium makes downward entropy fluctuations of size ΔS with relative frequency of order $e^{-\Delta S}$. Small fluctuations then vastly outnumber the single large one that a Big Bang requires, and a typical observer is a fluctuation rather than the product of a low-entropy past.

Carroll, Diachenko and Dulani [7] propose a way out. Following [6], they take the shifted energies to be commensurable, $E_k - E_0 = p_k \omega_*$ with $p_k \in \mathbb{Z}$. The evolution is then exactly periodic with period $T_* = 2\pi/\omega_*$, and its orbit is a closed curve in the torus of phases rather than a dense winding. Coarse-graining is a decomposition $\mathbb{C}^D = \bigoplus_a \mathcal{H}_a$ into macrostates of dimensions d_a with projectors Π_a. The quantum Boltzmann entropy is

$$S_{\mathrm{QB}}(t) = \sum_a p_a(t) \log d_a, \qquad p_a(t) = \langle \psi(t) | \Pi_a | \psi(t) \rangle. \tag{1}$$

Starting in the smallest macrostate, the authors argue that the bounce at $t = 0$ "can periodically repeat on a much shorter timescale than one would expect from $\exp(-\Delta S)$ statistics". They support this with a simulation ($D = 111$, macrostates of dimensions $1, 10, 100$ placed Haar-randomly relative to the energy basis, energies equal to random integers in $[1, 1000]$ times ω_*). Ordinary observers are counted once per period and Boltzmann brains at a fixed rate per unit time. This gives the criterion $T_* \le e^{10^{66}}$, and the authors conclude that a period in the range

$$10^{10}\,\mathrm{yr} \le T_* \le e^{10^{66}} \tag{2}$$

"should be completely allowed". Observers on different branches are to be weighted "by the amplitude-squared of the branches that they are on, ... so a standard expectation value in a single stochastic universe is what we want" [7, footnote 2].

We show that the argument fails. Throughout, $H = \sum_{j=1}^{L} E_j P_j$ is the spectral decomposition of the Hamiltonian into distinct eigenvalues $E_1 < \cdots < E_L$ with multiplicities $m_j = \operatorname{rank} P_j$. The initial state is a unit vector ψ_0 in the range of a macrostate projector Π of rank d. We write $q_j = \|P_j\psi_0\|^2$, $M = \#\{j : q_j > 0\}$, $\Delta E^2 = \sum_j q_j E_j^2 - (\sum_j q_j E_j)^2$, and

$$\bar\rho = \sum_j P_j|\psi_0\rangle\langle\psi_0|P_j, \qquad \bar p = \operatorname{Tr}(\bar\rho\,\Pi). \tag{3}$$

A *period* of the evolution is any $T > 0$ for which $\psi(T)$ is a multiple of ψ_0. Such a T exists exactly when the populated gaps are commensurable, and then every populated gap lies in $(2\pi/T)\mathbb{Z}$.

Theorem A (Spectral invariance)**.** *For every operator A the long-time average of $\langle\psi(t)|A|\psi(t)\rangle$ equals $\operatorname{Tr}(\bar\rho A)$. When the evolution has a period T, the average over any interval of length T already equals $\operatorname{Tr}(\bar\rho A)$. In particular, every Born-weighted rate of events depends on H only through its spectral projections. Every commensurable H with at least three distinct eigenvalues is a limit of incommensurable Hamiltonians with exactly the same Born-weighted rates.*

Theorem B (Weight of the minimal macrostate)**.** *One has $\sum_j q_j^2 \le \bar p \le \max_j \|P_j\Pi P_j\|$, and $\sum_j q_j^2 \ge 1/M$. If the macrostate decomposition is placed Haar-randomly relative to the eigenspaces, with ψ_0 correspondingly random in $\mathcal{H}_1$, then*

$$\mathbb{E}\,\bar p = \sum_{j=1}^{L} \frac{m_j(m_j+1) + (d-1)\,m_j(D-m_j)/(D-1)}{D(D+1)}. \tag{4}$$

For a nondegenerate spectrum this equals $(d+1)/(D+1)$, and in general $\mathbb{E}\,\bar p \le \sum_j (m_j/D)^2 + d/(D+1)$.

Theorem C (The period is not free)**.** *Let T be a period and suppose $\bar p \le 8/15$. Then*

$$T\,\Delta E \;\ge\; \frac{16\sqrt{2}}{15}\,\frac{1}{\bar p} \;>\; \frac{1.5}{\bar p}.$$

Moreover, if W is the spread of the populated energies, then $M \le 1 + WT/2\pi$, so the von Neumann and Rényi-2 entropies of $\bar\rho$ are at most $\log(1 + WT/2\pi)$.

Theorem A removes the mechanism. The only role of commensurability in [7] is to make the bounce recur at full weight. Under the Born weighting that the authors prescribe, a full-weight bounce once per period produces exactly the same expected numbers of every kind of observer, per unit time, as the partial near-recurrences of a nearby incommensurable spectrum. Theorems B and C show that nothing replaces the mechanism. In the ensemble used in [7], the minimal macrostate is occupied for the fraction $(d+1)/(D+1)$ of the time. That is the $e^{-\Delta S}$ law up to a factor at most 2, and bounces can recur no faster than $\Delta E\,\bar p/1.5$. Corollary 5.1 applies this to the range (2).

The facts behind Theorem A go back to von Neumann's dephasing argument [12] and underlie the modern treatment of equilibration [11, 9]. The state $\bar\rho$ is the diagonal ensemble, and its entropy is the diagonal entropy of [10]. What is new is their application to the proposal of [7], the exact Haar formula (4), and the period bound of Theorem C.

2. Dephasing and spectral invariance

Proof of Theorem A. Since $\psi(t) = \sum_j e^{-iE_jt}P_j\psi_0$,

$$\langle\psi(t)|A|\psi(t)\rangle = \sum_{j,l} e^{i(E_j-E_l)t}\langle P_j\psi_0|A|P_l\psi_0\rangle.$$

For $\nu \neq 0$, $\frac{1}{T}\int_{t_0}^{t_0+T} e^{i\nu t}\,\mathrm{d}t = e^{i\nu t_0}(e^{i\nu T}-1)/(i\nu T)$. This tends to 0 as $T \to \infty$, and it vanishes identically when $\nu \in (2\pi/T)\mathbb{Z} \setminus \{0\}$. Every off-diagonal frequency $E_j - E_l$ with $q_j q_l > 0$ is of this kind when T is a period, and terms with $q_j q_l = 0$ vanish. Only the diagonal $j = l$ survives, and it gives $\sum_j \langle P_j\psi_0|A|P_j\psi_0\rangle = \mathrm{Tr}(\bar\rho A)$.

A Born-weighted count of events of a given kind up to time τ has the form $N_O(\tau) = \int_0^\tau \langle\psi(t)|O|\psi(t)\rangle\,\mathrm{d}t$ for a positive operator O. Its rate $\lim N_O(\tau)/\tau = \mathrm{Tr}(\bar\rho O)$ depends on the projections P_j but not on the numbers E_j. Given a commensurable H, put $H_\varepsilon = \sum_j (E_j + \varepsilon\eta_j)P_j$ with $\eta_j = \sqrt{\pi_j}$ for distinct primes π_j. For small $\varepsilon > 0$ the order of the eigenvalues is unchanged. If $L \geq 3$, the ratio of the gaps $E_2 - E_1 + \varepsilon(\eta_2 - \eta_1)$ and $E_3 - E_1 + \varepsilon(\eta_3 - \eta_1)$ is a nonconstant Möbius function of ε, because $(\eta_2 - \eta_1)/(\eta_3 - \eta_1)$ is irrational by the linear independence of square roots of distinct primes over $\mathbb{Q}$; it is therefore irrational for all but countably many ε. So H_ε is incommensurable, $H_\varepsilon \to H$, and H_ε has the same projections and hence the same rates. □

Remark 2.1. The criterion behind (2) compares N_{OO} ordinary observers per period with $N_{\mathrm{BB}} = \Gamma_{\mathrm{BB}} T_* H_*^{-3}$ Boltzmann brains per period. Under Born weighting the first number is a rate $N_{\mathrm{OO}}/T_* = \mathrm{Tr}(\bar\rho\, O_{\mathrm{OO}})$ and the second is $\mathrm{Tr}(\bar\rho\, O_{\mathrm{BB}})$. By Theorem A their ratio is the same for H and for every H_ε, whose evolution has no period at all. The ratio therefore cannot be controlled by the length of the period. What it does depend on is the diagonal ensemble, and Theorems B and C show what the diagonal ensemble allows.

3. The weight of the minimal macrostate

Proof of the deterministic part of Theorem B. By (3), $\bar p = \sum_j \|\Pi P_j\psi_0\|^2$. Since $\Pi\psi_0 = \psi_0$, the Cauchy–Schwarz inequality gives $\|\Pi P_j\psi_0\| \geq |\langle\psi_0|\Pi P_j\psi_0\rangle| = |\langle\psi_0|P_j\psi_0\rangle| = q_j$. Hence $\bar p \geq \sum_j q_j^2 \geq (\sum_j q_j)^2/M = 1/M$. For the upper bound, $\|\Pi P_j\psi_0\|^2 = \langle P_j\psi_0|P_j\Pi P_j|P_j\psi_0\rangle \leq \|P_j\Pi P_j\|\, q_j$, and $\sum_j q_j = 1$. □

The lower bound $\sum_j q_j^2 = \mathrm{Tr}(\bar\rho^{\,2})$ is the time-averaged return probability. The upper bound says that the minimal macrostate can be occupied much more often than its share of Hilbert space only if it overlaps some eigenspace of H much more than its share. That is a statement about the position of $\mathcal{H}_1$ relative to the eigenspaces, not about the eigenvalues.

For the random part we use the model of [7]. Fix the eigenspaces, fix an orthonormal basis $e_1, \ldots, e_D$ and a partition of $\{1, \ldots, D\}$ into blocks B_a of sizes d_a with $1 \in B_1$. Let U be Haar-distributed on $\mathrm{U}(D)$, and put $\mathcal{H}_a = U\,\mathrm{span}\{e_i : i \in B_a\}$ and $\psi_0 = Ue_1$. Because the Haar measure is invariant under unitaries of $\mathcal{H}_1$, this gives the same law as an arbitrary rule for choosing ψ_0 in $\mathcal{H}_1$ that is unitarily covariant.

Lemma 3.1. *Let u be uniform on the unit sphere of $\mathbb{C}^D$ and P a projector of rank m. Then $q = \langle u|P|u\rangle$ satisfies $\mathbb{E}q = m/D$ and $\mathbb{E}q^2 = m(m+1)/(D(D+1))$. If v is uniform on the unit sphere of $u^\perp$, then $\mathbb{E}\,|\langle v|P|u\rangle|^2 = m(D-m)/(D(D+1)(D-1))$.*

Proof. Write $u = g/\|g\|$ with g a standard complex Gaussian vector, and take P diagonal, which is allowed by unitary invariance. The variables $|g_i|^2$ are independent and exponential with mean one. So $q = X/(X+Y)$ with independent $X \sim \Gamma(m)$ and $Y \sim \Gamma(D-m)$, that is, q has the $\mathrm{Beta}(m, D-m)$ law, whose first two moments are those stated. Given u, the vector v is invariant in law under unitaries fixing u, so $\mathbb{E}[|v\rangle\langle v| \mid u] = (I - |u\rangle\langle u|)/(D-1)$. Hence

$$\mathbb{E}\big[|\langle v|Pu\rangle|^2 \mid u\big] = \frac{\|Pu\|^2 - |\langle u|Pu\rangle|^2}{D-1} = \frac{q - q^2}{D-1},$$

and taking expectations gives $\big(m/D - m(m+1)/(D(D+1))\big)/(D-1) = m(D-m)/(D(D+1)(D-1))$. □

Proof of (4). Since $\Pi_1 = \sum_{i\in B_1} |Ue_i\rangle\langle Ue_i|$,

$$\bar{p} = \sum_j \sum_{i\in B_1} |\langle Ue_i|P_j|Ue_1\rangle|^2.$$

The vector $u = Ue_1$ is uniform on the sphere, and for $i \neq 1$ the vector Ue_i is uniform on the sphere of $u^\perp$. Lemma 3.1 gives $m_j(m_j+1)/(D(D+1))$ for $i = 1$ and $m_j(D-m_j)/(D(D+1)(D-1))$ for each of the $d-1$ other indices. Summing over j gives (4). If every $m_j = 1$, then $L = D$ and each summand is $(d+1)/(D(D+1))$. In general $(D-m_j)/(D-1) \leq 1$ and $\sum_j m_j = D$, so $\mathbb{E}\bar{p} \leq (\sum_j m_j^2 + dD)/(D(D+1)) \leq \sum_j (m_j/D)^2 + d/(D+1)$. □

For the other macrostates the same computation gives $\mathbb{E}\,\mathrm{Tr}(\bar{\rho}\,\Pi_a) = d_a/(D+1)$ for $a \neq 1$ in the nondegenerate case, and the weights sum to one. So in the ensemble of [7] the diagonal ensemble gives every macrostate its share of Hilbert space. The macrostate holding the initial state gets one extra unit, (d_1+1) in place of d_1, and no more.

Remark 3.2 (The simulation of [7])**.** There $d_1 = 1$ and the $D = 111$ energies are independent uniform integers in $[1, 1000]$, so ties occur. With $d = 1$, formula (4) reads $\mathbb{E}\bar{p} = (D + \sum_j m_j^2)/(D(D+1))$. Since $\mathbb{E}\sum_j m_j^2 = \sum_{k,k'} \mathbb{P}(E_k = E_{k'}) = D + D(D-1)/1000$, we get $\mathbb{E}\bar{p} = 234.21/12432 \approx 1/53$. This is about twice $d_1/D = 1/111$, as Theorem B predicts. The weight is not concentrated at the bounce. Figure 1 shows one realisation of this set-up, where the window $|t| \leq 3/\Delta E$ around the bounce carries 6% of the time-averaged weight of $\mathcal{H}_1$. The remaining 94% comes from the small fluctuations of the equilibrium phase. The figure is an illustration only and is not used in any proof.

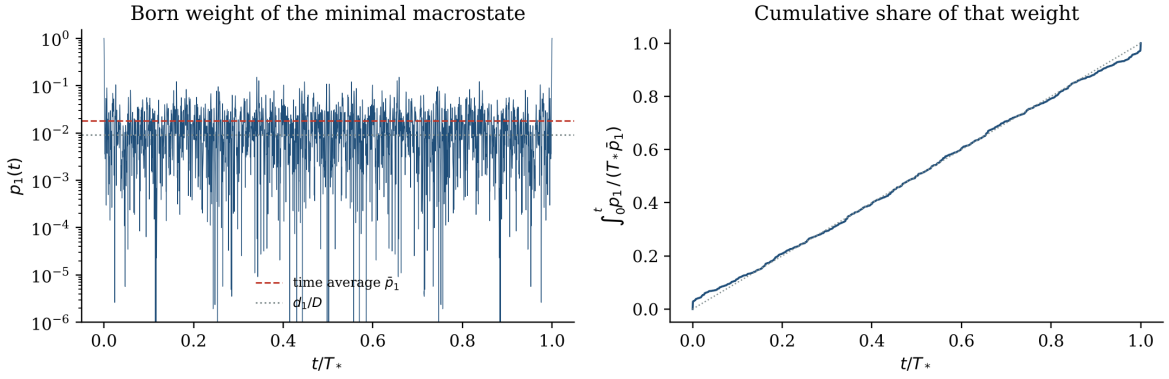


FIGURE 1. One realisation of the set-up of [7]: $D = 111$, $d_a = 1, 10, 100$, energies uniform integers in $[1, 1000]$ (period $T_* = 2\pi/\omega_*$). Left: $p_1(t)$ on a logarithmic scale, with its time average $\bar{p}_1 \approx 0.018$ and the share d_1/D. Right: cumulative fraction of $\int p_1$ over one period. The bounces at $t = 0$ and $t = T_*$ add a visible but small step to an otherwise uniform growth.

4. The period bound

Lemma 4.1. *For all real t, $|\langle\psi_0|\psi(t)\rangle| \geq 1 - \frac{1}{2}\Delta E^2 t^2$, and therefore $p(t) = \|\Pi\psi(t)\|^2 \geq \big(1 - \frac{1}{2}\Delta E^2 t^2\big)^2$ whenever $|t| \leq \sqrt{2}/\Delta E$.*

Proof. Let $\bar{E} = \sum_j q_j E_j$. Then $|\langle\psi_0|\psi(t)\rangle| = |\sum_j q_j e^{-i(E_j-\bar{E})t}| \geq \sum_j q_j \cos((E_j - \bar{E})t)$. Using $\cos x \geq 1 - x^2/2$ bounds this below by $1 - \frac{1}{2}\Delta E^2 t^2$. Since $\Pi\psi_0 = \psi_0$, we have $|\langle\psi_0|\psi(t)\rangle| = |\langle\psi_0|\Pi\psi(t)\rangle| \leq \|\Pi\psi(t)\|$. □

This is an elementary form of the Mandelstam–Tamm speed limit. Its only use here is to show that a bounce lasts for a time of order $1/\Delta E$.

Proof of Theorem C. By Theorem A, $\bar{p} = \frac{1}{T}\int_{-T/2}^{T/2} p(t)\,\mathrm{d}t$. Put $s = T\Delta E$ and $g(x) = (1-x^2/2)^2$ for $|x| \leq \sqrt{2}$.

If $s < 2\sqrt{2}$, Lemma 4.1 applies on the whole interval and gives $\bar{p} \geq \frac{1}{s}\int_{-s/2}^{s/2} g(x)\,\mathrm{d}x$. The right side decreases in s, since g decreases in $|x|$, and at $s = 2\sqrt{2}$ it equals $\frac{1}{2\sqrt{2}}\int_{-\sqrt{2}}^{\sqrt{2}} g = \frac{1}{2\sqrt{2}} \cdot \frac{16\sqrt{2}}{15} = \frac{8}{15}$. This would force $\bar{p} > 8/15$, so the hypothesis gives $s \geq 2\sqrt{2}$. In that case we integrate only over $|t| \leq \sqrt{2}/\Delta E$ and use $p \geq 0$ elsewhere:

$$\bar{p} \;\geq\; \frac{1}{T\Delta E}\int_{-\sqrt{2}}^{\sqrt{2}} \Big(1 - \frac{x^2}{2}\Big)^2 \,\mathrm{d}x = \frac{16\sqrt{2}}{15}\,\frac{1}{T\Delta E},$$

which is the first claim.

For the second claim, every populated gap $E_j - E_l$ lies in $(2\pi/T)\mathbb{Z} \cap [-W, W]$. So the populated energies are distinct points of an arithmetic progression of step $2\pi/T$ in an interval of length W, and there are at most $1 + WT/2\pi$ of them. The operator $\bar{\rho}$ has the orthogonal eigenvectors $P_j\psi_0$ with eigenvalues q_j. It therefore has rank M, and both its von Neumann entropy and its Rényi-2 entropy $-\log\sum_j q_j^2$ are at most $\log M$. □

Theorem C is a rigorous version of the statement that bounces recur at most at the rate their Born weight allows. In the nondegenerate case with macrostates in general position, Theorem B and Markov's inequality show that, for every $\delta \in (0,1)$, with probability at least $1-\delta$ we have $\bar{p} \leq (d+1)/(\delta(D+1))$. On that event, simultaneously for *every* choice of distinct eigenvalues with these eigenvectors that makes the evolution periodic,

$$T\,\Delta E \;\geq\; 1.5\,\delta\,\frac{D+1}{d+1} \qquad \text{provided } \frac{d+1}{\delta(D+1)} \leq \frac{8}{15}. \tag{5}$$

The period thus grows like $e^{\Delta S}$ with $\Delta S = \log(D/d)$, contrary to the statement in [7] that T_* has no dependence on the size of Hilbert space. This holds even though the evolution is exactly periodic. The authors' heuristic for the time to enter a macrostate, $e^{c(D-d)}/\delta E$, also overstates the scale. The correct Born-weighted scale is $e^{\Delta S}/\Delta E$, exponential in the entropy rather than in the dimension.

5. Consequences for the proposed range of periods

Corollary 5.1. *Suppose a periodic finite quantum universe has period $T \leq e^{10^{66}}$, with its populated energy spread W and energy uncertainty ΔE at most $e^{O(1)}$ in Planck units. Suppose the initial state lies in a macrostate of dimension d, and the dimension D is so large that $d/D \leq e^{-10^{100}}$. Then:*

1. *the diagonal ensemble has von Neumann entropy at most $10^{66} + O(1)$;*
2. *$\bar{p} \geq e^{-10^{66}-O(1)}$, so for Haar-random macrostates the bound $\mathbb{E}\bar{p} \leq \sum_j (m_j/D)^2 + d/(D+1)$ forces either $\sum_j (m_j/D)^2 \geq \delta\, e^{-10^{66}-O(1)}$ or an event of probability less than δ.*

Proof. Part (1) is the second half of Theorem C, since $\log(1+WT/2\pi) \le 10^{66}+O(1)$. For part (2), Theorem C gives $\bar{p} \ge 1.5/(T\Delta E) \ge e^{-10^{66}-O(1)}$ when $\bar{p} \le 8/15$, and trivially otherwise. By Markov's inequality, $\mathbb{P}(\bar{p} \ge x) \le (\sum_j (m_j/D)^2 + d/(D+1))/x$. Taking $x = e^{-10^{66}-O(1)}$ and using $d/(D+1) \le e^{-10^{100}}$ gives the dichotomy. □

The range (2) is therefore reachable in only two ways, and commensurability plays no part in either.

The first is fine-tuning. The low-entropy macrostate must sit relative to the energy eigenspaces in a position whose Haar probability is double-exponentially small. In the de Sitter setting of [7], with $D \sim e^{10^{122}}$, that probability is of order $e^{10^{66}-10^{122}}$.

The second is a spectrum concentrated on at most about $e^{10^{66}}$ effective levels, which is the degeneracy the authors invoke. By part (1), the state of the universe then explores a subspace of dimension at most about $e^{10^{66}}$, the dimension equivalent of the entropy $S_{\mathrm{BB}} \approx 10^{66}$ of one Boltzmann brain used in [7]. Its equilibrium carries at most that much diagonal entropy, far from the 10^{122} of the de Sitter phase that the scenario identifies with equilibrium. The Boltzmann brain rate $\Gamma_{\mathrm{BB}} \sim e^{-M_{\mathrm{BB}}/T_{\mathrm{dS}}}$ that enters the criterion presupposes thermal fluctuations of a de Sitter patch with entropy 10^{122}. A dynamics confined to a subspace of dimension $e^{10^{66}}$ does not supply that estimate.

Either way, the ratio of ordinary observers to fluctuations is fixed by the diagonal ensemble (Theorem A). It is the same as for a nearby incommensurable universe with no recurrences at all.

Remark 5.2 (What is correct in [7])**.** The exact periodicity for commensurable gaps, the non-ergodicity of the orbit on the torus of phases, and the criterion for the symmetry $S_{\mathrm{QB}}(t) = S_{\mathrm{QB}}(-t)$ (reality of the Fourier coefficients of each p_a) are all correct. The discretisation of [6] is also correct. With $N = \mathrm{lcm}(p_k)$ and $\gcd(p_k) = 1$, the minimal period in steps is $\mathrm{lcm}_k(N/p_k) = N/\gcd_k(p_k) = N$. The error lies in the inference from periodicity to the suppression of Boltzmann fluctuations.

The diagonal-ensemble viewpoint used here sits beside the author's earlier work on operator-theoretic and geometric structures [2, 3, 4].

Declarations

Data availability. No data were generated or analysed. All results are proved in the text. The LaTeX source and the script that draws Figure 1, which is not used in any proof, are archived at `https://doi.org/10.5281/zenodo.ZENODOID`.

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